Euclidean Geometry In Mathematical Olympiads

Lemma 1.44 (Three Tangents)

Chapter 1 fgm

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277546
1607 posts

Mar 13, 2018, 3:08 PM • 3 #1

Lemma 1.44 (Three Tangents)

Let ABC be an acute triangle. Let BE and CF be the altitudes of $\triangle ABC$, and denote by M the midpoint of BC. Prove that ME, MF, and the line through A parallel to BC are all tangents to (AEF).

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Mar 13, 2018, 3:12 PM • 4 #2

Solution (Part 1)

Let H be the orthocenter of $\triangle ABC$. First observe $AEHF$ is cyclic.The circumcenter of $\triangle AEH$ lies on the midpoint of $\triangle AEH$, namely O. Hence, A, O, and H are colinear By definition, $AH \perp BC$, so the line parallel to BC through A is perpendicular to AO.Hence, it is tangent to $(AEF)\blacksquare$.

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Solution (Part 2)

Notice that $FM = BM$ (since $\triangle BFC$ is right), so $\angle BFM = \angle ABC$. Since $\angle AFE = \angle ACB$ we get $\angle EFM = \angle BAC$. Since $\angle FAE = \angle BAC$ we see that FM is tangent to (AFE). An identical argument works for EM. $\blacksquare$

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Drunken_Master
327 posts

Mar 14, 2018, 4:11 AM • 6 #4

Solution

Let H be the orthocenter, since $\angle HAE = 90^\circ - \angle C = 90^\circ - \angle MEC = \angle MEH$, ME is tangent, and similarly MF is also tangent. Since AH is diameter, and $AH \perp BC$, line parallel to BC through A is tangent to (AEF).

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Durjoy1729
221 posts

Jun 22, 2018, 3:12 PM • 5 #5

Consider nine point circle and defined the Orthocenter H.Now observe that AH is perpendicular to BC at D.Points D, E, F, M and midpoint of AH namely center of (ADE) called O exists on nine point circle and MO is the diameter of the circle. $\angle OEM = 90^\circ, \angle OFM = 90^\circ$

This post has been edited 2 times. Last edited by Durjoy1729, Jun 22, 2018, 3:14 PM

proshi
2141 posts

Dec 27, 2018, 6:45 PM • 2 #6

> *realquarterb wrote:*
> Solution (Part 2)

Why?

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Malstern
198 posts

Jan 3, 2019, 1:48 AM • 4 #7

> *realquarterb wrote:*
> Solution (Part 2)

Here CI = CG , but neither is tangent to the circle.

Attachments: